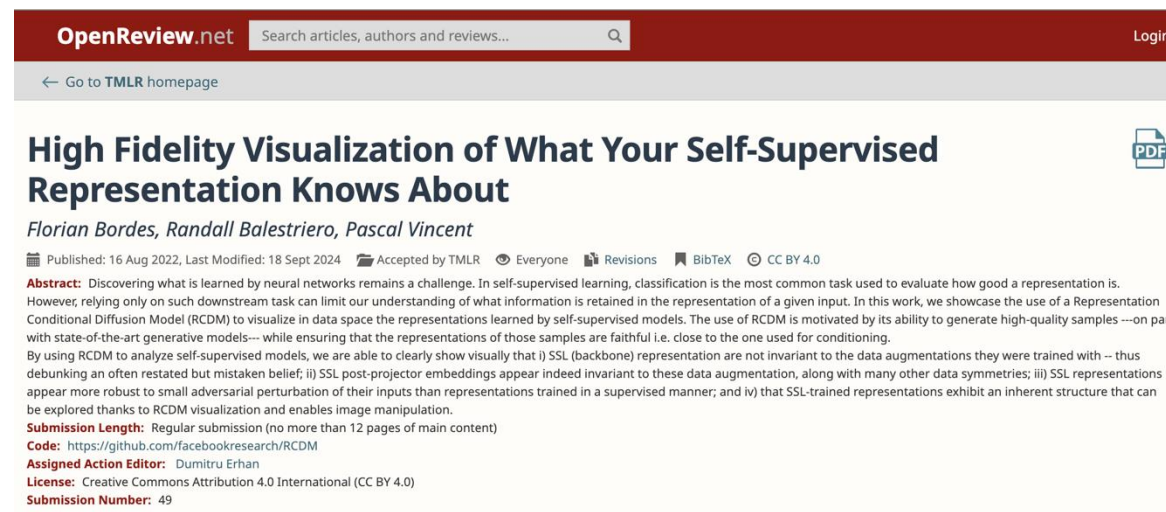


# High Fidelity Visualization of What Your Self-Supervised Representation Knows About (RCDM)

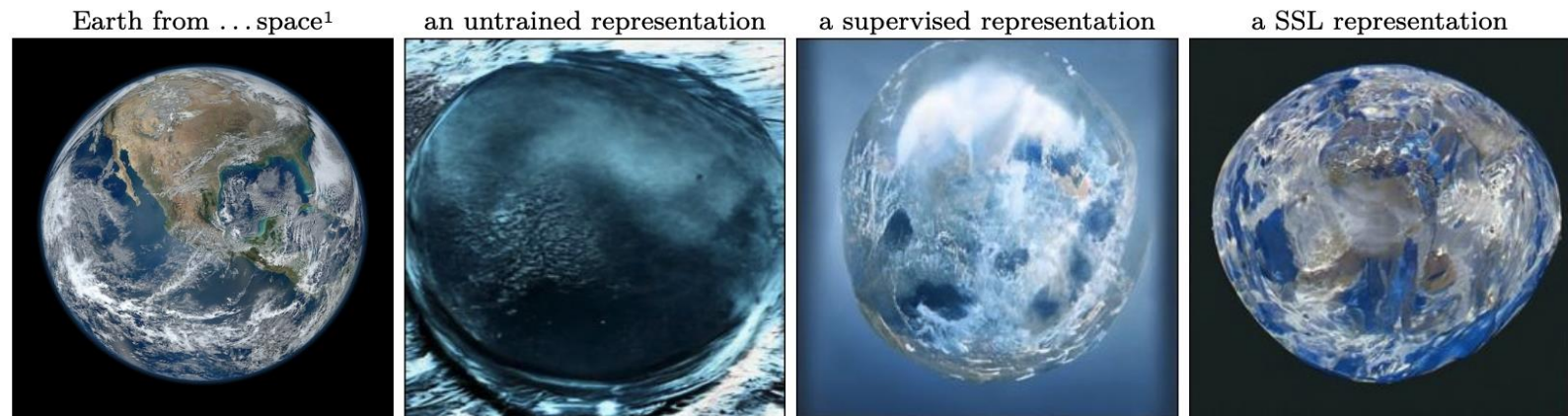
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- Introduction
- Related Work
- High-Fidelity Conditioning with Diffusion Models
- Visual Analysis of Representations Learned by Self-Supervised Model
- Conclusion
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<sup>1</sup>We use representations of the real picture of Earth on the left (source: NASA) as conditioning for RCDM. We show samples (resolution  $256 \times 256$ ) in cases where the representations (2048-dimensions) were obtained respectively with a random initialized ResNet50, a supervised-trained one, and a SSL-trained one. More samples in Fig. 36.

- SSL feature / embedding의 품질을 평가하는 법
  - w/ Label – Linear prob. / KNN / Full Fine Tuning ... etc
  - w/o Label – RankMe / LiDAR ... etc
  - Visual – **RCDM**
- **RQ. What information does the learned representation contain?**
  - SSL은 보통 transform, corruption 등을 통해 invariance 한지에 대하여 학습함.
    - 어떻게 invariance한지 확인할까?
  - SSL에서 projector를 쓰는 것과 backbone만 쓰는 것은 정확도 차이가 10~30% 향상된다는데 ...
    - 그럼 Backbone에는 뭐가 있고 projector에는 뭐가 있길래? 그렇게 차이가 날까?

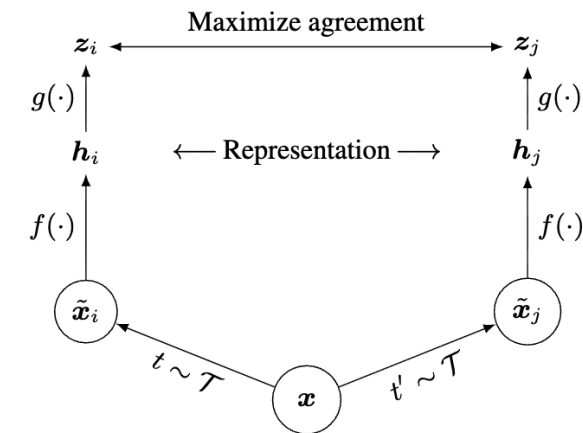


Figure 2. A simple framework for contrastive learning of visual representations. Two separate data augmentation operators are sampled from the same family of augmentations ( $t \sim \mathcal{T}$  and  $t' \sim \mathcal{T}$ ) and applied to each data example to obtain two correlated views. A base encoder network  $f(\cdot)$  and a projection head  $g(\cdot)$  are trained to maximize agreement using a contrastive loss. After training is completed, we throw away the projection head  $g(\cdot)$  and use encoder  $f(\cdot)$  and representation  $h$  for downstream tasks.

## Contribution

- Diffusion Model이 SSL large vector representation을 conditioning으로 받을 수 있음을 증명
  - 생성 이미지가 high-quality인 동시에 representation-faithful
    - 생성된 이미지를 다시 인코딩하면 원래 표현과 유사한 벡터가 나옴
- 같은 표현으로 여러 번 샘플링 하여 정성적으로 분석하는 데 잠재적인 유용성을 보임
  - 항상 동일한 부분 = 표현에 인코딩된 정보 매번 달라지는 부분 = 버려진 정보

## Observations

1. 불변성: SSL projector > Supervised > SSL backbone
2. SSL은 배경·스타일 보존, Supervised는 무시
3. SSL도 객체 크기, 색상, 배경 색조는 보존됨
4. Supervised가 adversarial attack에 더 취약함
5. SSL 표현을 전경/배경으로 분리해 조작 가능

## Visualization methods

- 기존 연구들은 신경망이 학습한 내용을 시각화하기 위해 **그래디언트 기반 기법**을 주로 사용했다. 특정 뉴런의 활성화를 최대화하거나, 네트워크를 "역전(invert)"시키는 방식으로 각 레이어에서 학습된 내용을 시각화했다.
  - 결정론적인 단점이 있음.

## Generative models

- VAE, GAN, Flow, Diffusion 등 다양한 생성 모델이 존재
- SSL처럼 **풍부한 representation을 conditioning으로 쓸 때의 핵심 난제**는, 하나의 conditioning 벡터  $h$ 에 대응하는 학습 데이터  $x$ 가 단 하나뿐이라는 것.
  - 클래스 레이블로 conditioning하면 같은 클래스 이미지가 수천 장이지만, SSL 표현은 그렇지 않음.
- 기존 방법은 레이블이 필요하거나 고해상도 이미지에 부적합 ( $32 \times 32$  수준).
- RCDM은 전부 해결

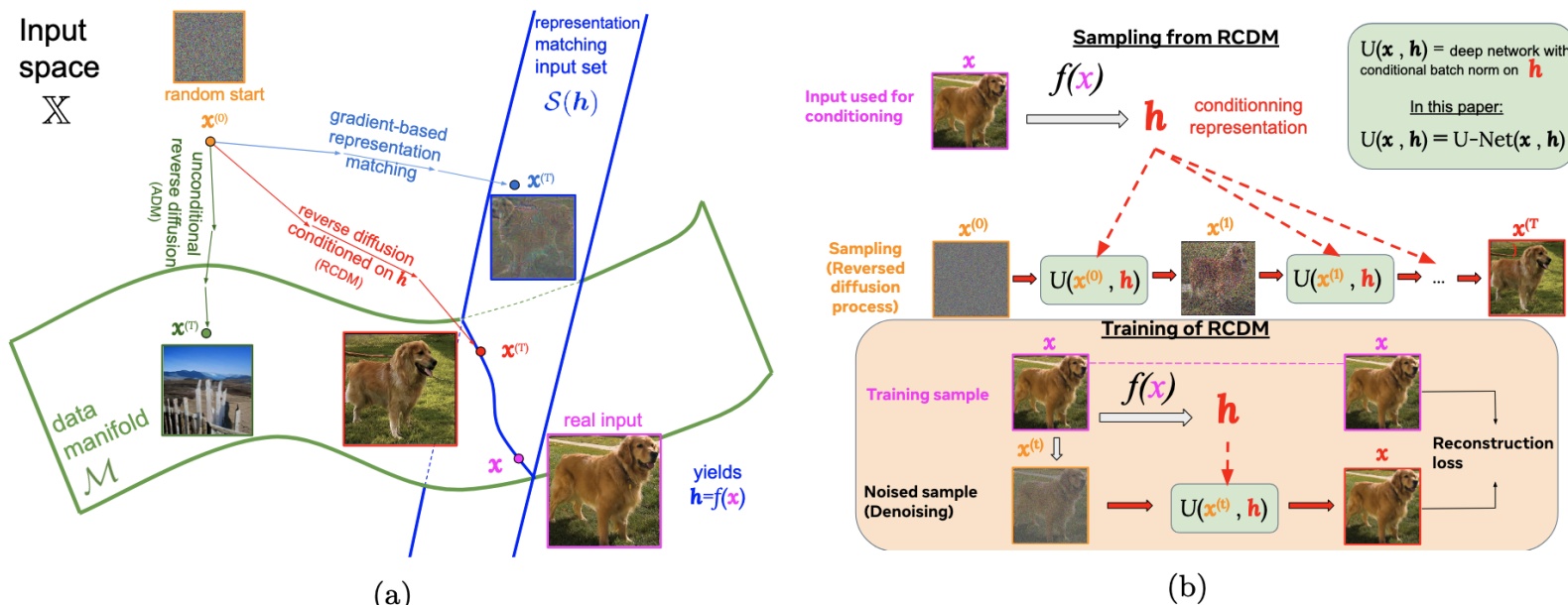


Figure 6: (a) Illustration of considered image generation methods. A real input  $\mathbf{x}$  yields representation  $\mathbf{h}$ . All methods start from a random noise image  $\mathbf{x}^{(0)}$ . Gradient-based representation matching (light blue arrows) will move it towards  $\mathcal{S}(\mathbf{h})$  i.e. until its representation matches  $\mathbf{h}$ , but won't land on the data-manifold  $\mathcal{M}$ . Unconditional reverse diffusion (ADM model, green arrows) will move it towards the data manifold. Our representation-conditioned diffusion model (RCDM, red arrows) will move it towards  $\mathcal{M} \cap \mathcal{S}(\mathbf{h})$ , yielding a different natural-looking image with the same given representation. (b) Representation-Conditioned Diffusion Model (RCDM). From a diffusion process that progressively corrupts an image, the model learns the reverse process by predicting the noise that it should remove at each step. We also add as conditioning a vector  $\mathbf{h}$ , which is the representation given by a SSL or supervised model for a given image  $\mathbf{x}$ . Thus, the network is trained explicitly to denoise towards a specific example given the corresponding conditioning. The diffusion model used is the same as the one presented by Dhariwal & Nichol (2021) with the exception of the conditioning on the representations.

## Ablated Diffusion Model (ADM) 기반

- $h = f(x)$ 을 잘 Conditioning하기 위해 Group normalization을 IC-GAN의 conditional batch normalization을 사용함.
- $h$ 를 512차원 벡터로 압축하여 각 Residual Block의 Conditional Batch Normalization에 입력하는 식

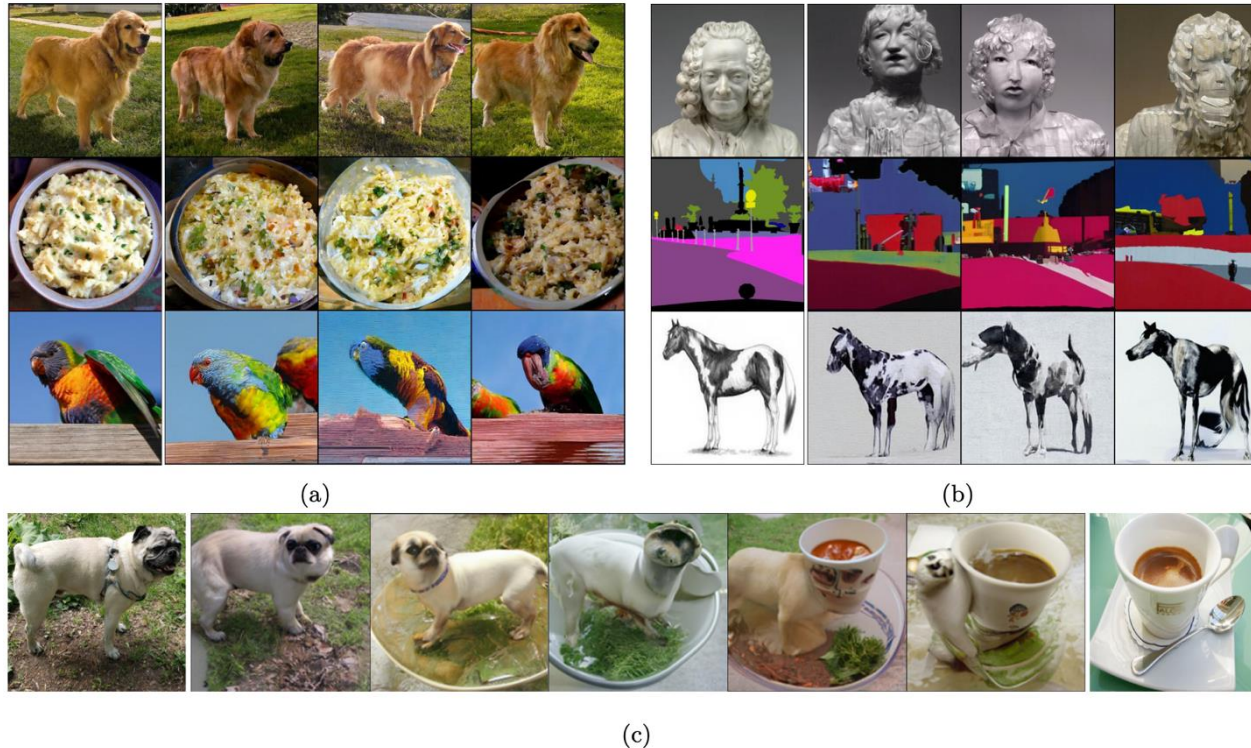
$$\text{Batch Normalization: } \hat{x} = \frac{x - \mu}{\sigma} \cdot \gamma + \beta$$

$$\text{Conditional Batch Normalization: } \hat{x} = \frac{x - \mu}{\sigma} \cdot \gamma(h) + \beta(h)$$

- ADM은 classifier-guidance로 label 정보를 조건으로 주입하는 것이 가능한데 이를 없앴으므로 학습 도중 label 정보를 조건으로 사용하지 않음.
- SSL은 freezing하고 훈련에 사용하지 않음.

## 1<sup>st</sup>: condition에 가까운 realistic-looking images를 만드는 능력이 되는가?

- DINO로 학습된 Resnet 50 backbone으로 (2048D feature) IN1k에 학습 후
  - (a) IN1k validation set condition (in distribution)
  - (b) OOD (RCDM이 단순히 IN1k를 암기한 게 아님을 증명)
  - (c) IN1k validation set interpolation (unlikely mix of factors에 해당하는 표현에서도 해석 가능한 이미지를 생성)



## 2<sup>nd</sup>: 생성된 이미지를 다시 인코딩하면, 원래 conditioning 표현이 가장 가까운 이웃인가?

- IN1k validation 이미지  $x$  선택
- SSL 인코더로 표현  $h = f(x)$  추출
- $h$  를 conditioning으로 RCDM에서 이미지 생성
- 생성된 이미지를 다시 인코더에 넣어  $h'$  추출
- $h'$ 의 가장 가까운 이웃이 원래  $h$  인지 확인

| Model                             | ↓Mean rank | ↑MRR |
|-----------------------------------|------------|------|
| Dino (Caron et al., 2021)         | 1.00       | 0.99 |
| Swav (Caron et al., 2020)         | 1.01       | 0.99 |
| SimCLR (Chen et al., 2020)        | 1.16       | 0.97 |
| Barlow T. (Zbontar et al., 2021)) | 1.00       | 0.99 |
| Supervised                        | 5.65       | 0.69 |

- **Mean Rank (낮을수록 좋음)**
  - IN1k validation(n=50K) representation들 사이에서  $h'$ 와 원래  $h$ 가 몇 번째로 가까운가?
- **MRR (Mean Reciprocal Rank, 높을수록 좋음)**
  - Rank의 역수 평균
    - Rank=1이면 MRR=1.0, Rank=2이면 MRR=0.5

## 3<sup>rd</sup> : Backbone과 projector는 어떤 차이가 있을까?

- Backbone
  - 생성 이미지들이 배경, 스타일, 객체 세부사항까지 일관되게 유지
  - 더 많은 정보를 가지고 있음
- Projector
  - 생성 이미지들이 전역적(global) 정보만 공유, 세부사항은 제각각

## 4<sup>th</sup> : 생성 품질은 어떨까?

| Method                                 | Res. | ↓FID | ↑IS        |
|----------------------------------------|------|------|------------|
| ADM (Dhariwal & Nichol, 2021)          | 256  | 26.8 | 34.5 ± 1.8 |
| IC-GAN (Casanova et al., 2021))        | 256  | 20.8 | 51.3 ± 2.2 |
| IC-GAN (Casanova et al., 2021) w/ KDE* | 256  | 21.6 | 38.6 ± 1.1 |
| <b>RCDM w/ KDE* (ours)</b>             | 256  | 19.0 | 51.9 ± 2.6 |

| Model           | SimCLR Trunk | SimCLR Head | Dino Trunk | Dino Head | Barlow T. Trunk | Barlow T. Head | VicReg Trunk | VicReg Head |
|-----------------|--------------|-------------|------------|-----------|-----------------|----------------|--------------|-------------|
| <b>Val acc.</b> | 69.1 %       | 61.2 %      | 74.8 %     | 64.9 %    | 72.6 %          | 62.9 %         | 72.3 %       | 62.2 %      |

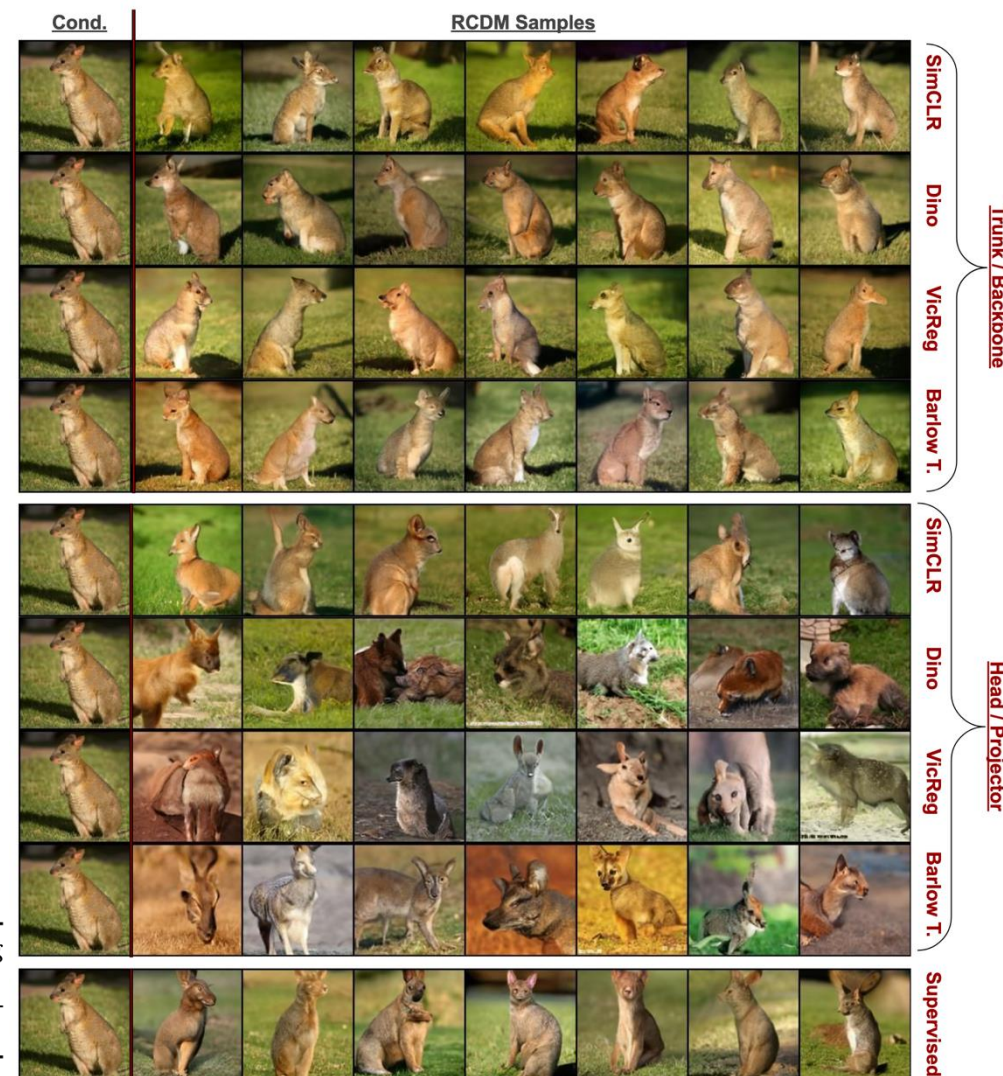
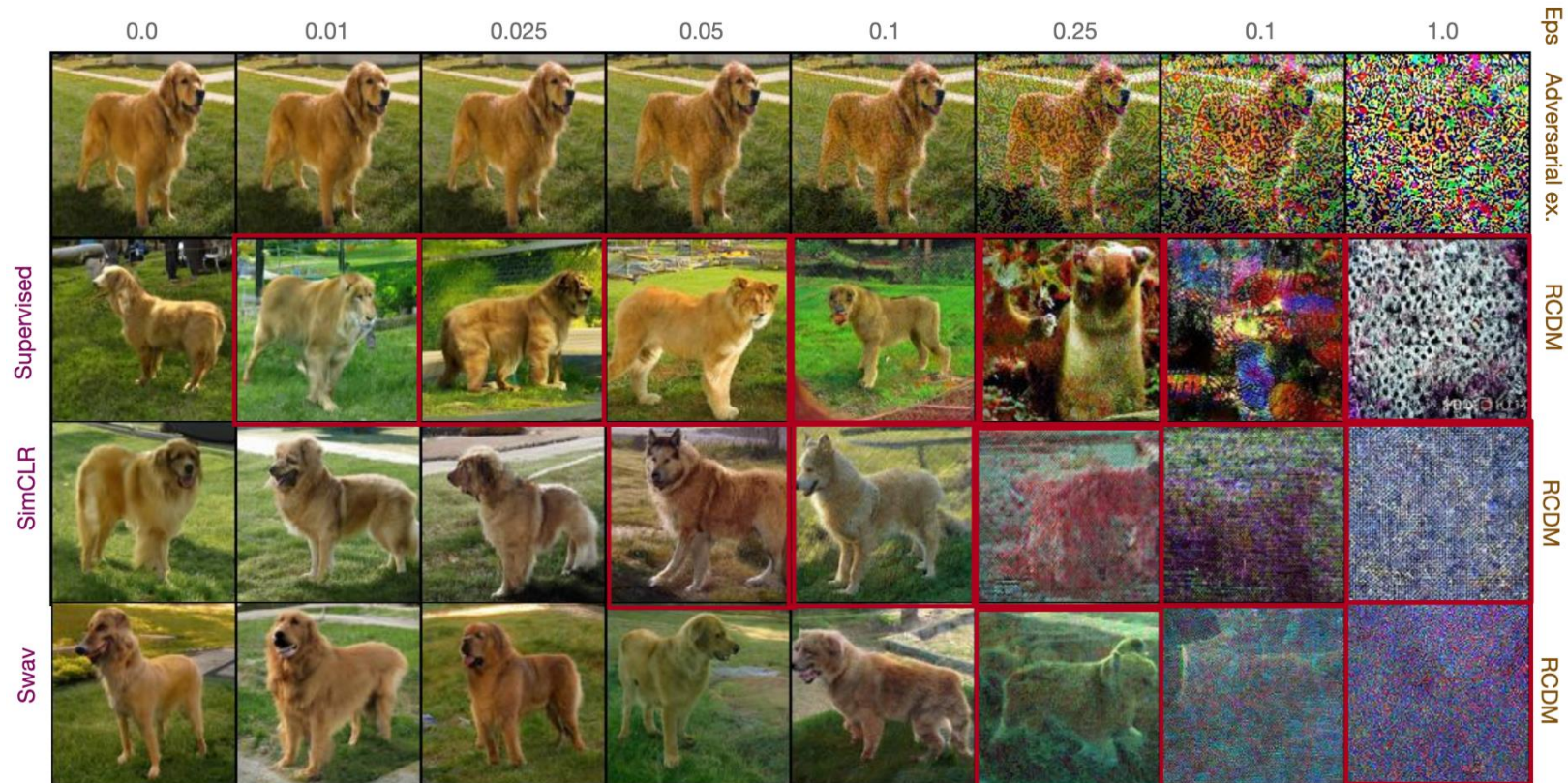


Table a): ImageNet linear probe validation accuracy on representation given by various SSL models. We observe an accuracy gap between the linear probes at the trunk level and the linear probes trained at the head level of around 10 percentage point of accuracy.

## Are Self-Supervised Representations Really Invariant to Data-Augmentations?

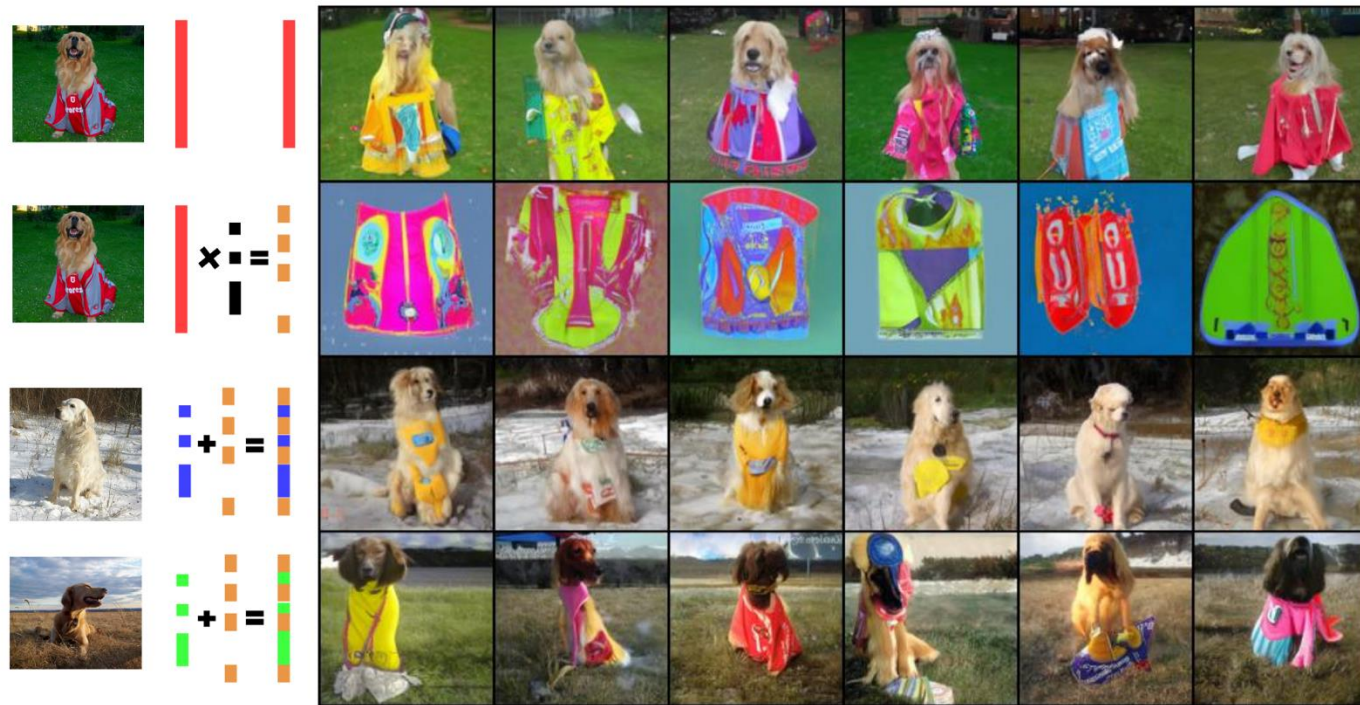


## Self-Supervised and Supervised Models See Adversarial Examples Differently



## Self-Supervised Representations Locally Encode background and Object on Different Dimensions

- conditioning 이미지의 nearest neighbor들을 찾음
- 이웃들 간에 공통으로 non-zero인 차원 = "공유 정보" 차원
- 이 차원들을 0으로 마스킹 → RCDM으로 생성
- 또는 다른 이미지의 값으로 교체 → RCDM으로 생성

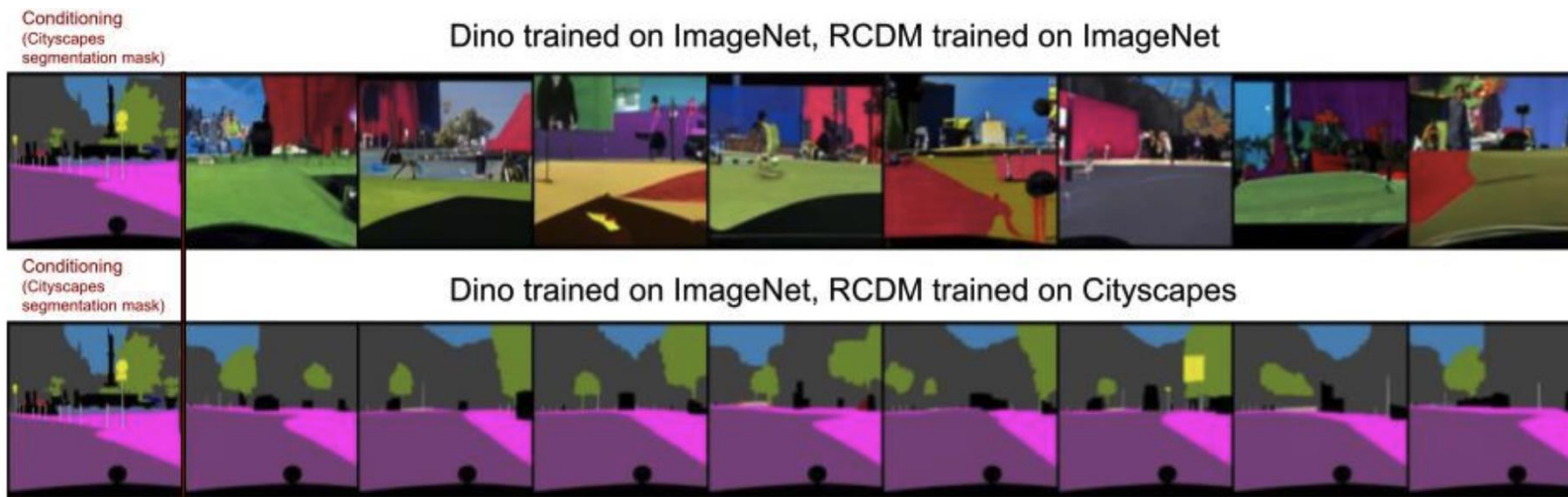


■ zero mask of most common indices where dim of representation is non zero

■ Least common dim of ■ where dim of representation is non zero

- RCDM을 통해 레이블 없이 생성 모델 하나로 SSL representation 안에 무엇이 있는지 직접 볼 수 있음
- RCDM을 통해 ...
  - 고품질 생성 (FID) + 표현 충실도 (Mean Rank  $\approx 1$ ) 확인
  - SSL vs. Supervised 표현의 차이를 다각도로 탐색
    - Invariance 차이
      - SSL과 Supervised가 각 augmentation에 다르게 반응함을 시각적으로 확인
    - Representation 차원의 구조
      - SSL 표현 공간에서 배경/객체가 다른 차원에 분리 인코딩됨  $\rightarrow$  조작 가능
    - Adversarial 민감도 차이
      - Supervised가 SSL보다 adversarial noise에 훨씬 취약

- OOD 이미지에서 생성이 실패할 때, 실패 원인이 어디에 있는가?
  - SSL 표현이 OOD 이미지를 충분히 담지 못해서?
  - RCDM이 OOD 이미지를 복원하지 못해서?



- DINO는 ImageNet만 학습했어도, Cityscapes 이미지의 표현을 충분히 담고 있었음
- 문제는 RCDM이 Cityscapes 스타일의 이미지를 생성하는 법을 몰랐던 것

## - Backbone vs Projector (Barlow Twins)

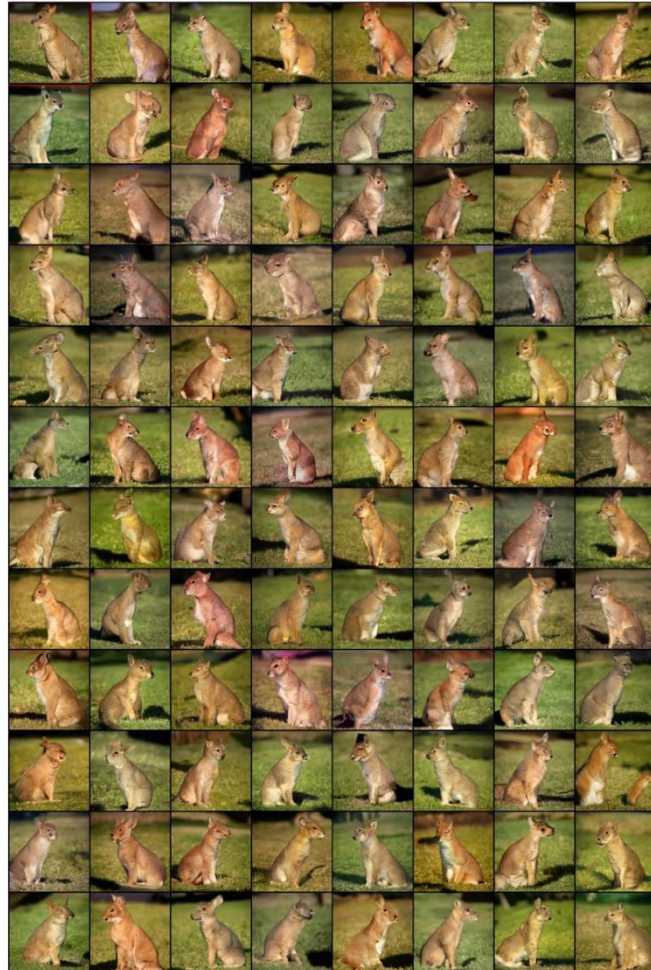


Figure 18: Visual analysis of the variance of the generated samples for a specific image when using the trunk/backbone of a Barlow Twins encoder. The first image (in red) is the one used as conditioning.



Figure 19: Visual analysis of the variance of the generated samples for a specific image when using the projector/head of a Barlow Twins encoder. The first image (in red) is the one used as conditioning.

## - Squared Euclidean distances Backbone (DINO) vs Projector (DINO)

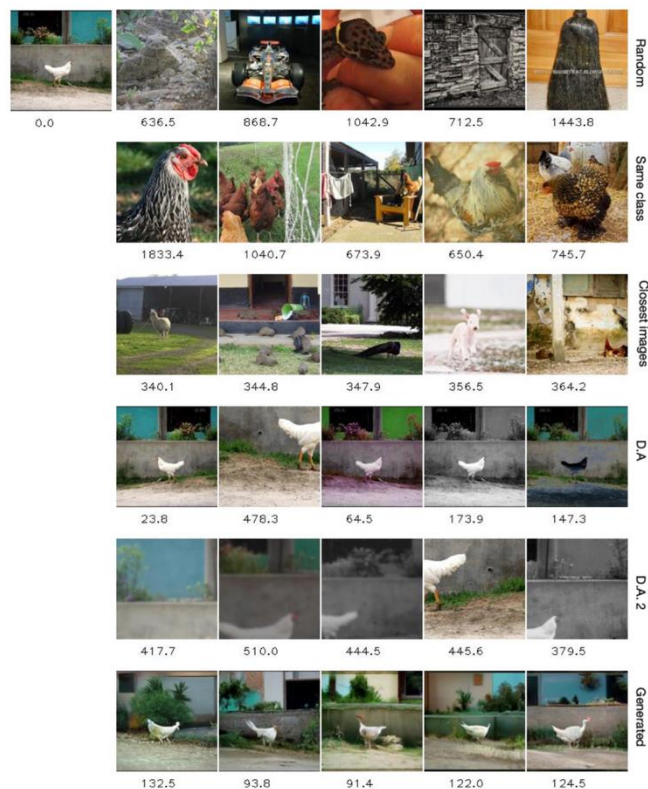


Figure 21: **Squared Euclidean distances in the DINO representation space.** We show the squared euclidean distance between the conditioning image on the leftmost column on first row and different images to get an insight about how close the samples generated by the diffusion model stay close to the representation used as conditioning. The distances with the conditioning is printed below each images. On the first row, we show random images from the ImageNet validation data. On the second row, we take random validation examples belonging to the same class as the conditioning. On the third row, we find the closest training neighbors of the conditioning in the representation space. On forth row, D.A. means Data Augmentation which consist in horizontal flip, CenterCrop, ColorJitter, GrayScale and solarization. On fifth row (D.A. 2), we use the random data Augmentation used in the paper of (Caron et al., 2020; 2021). On the last row, we show the generated samples from our conditional diffusion model that use **DINO representation**. The samples produces by our model are much closer to the conditioning than other images.

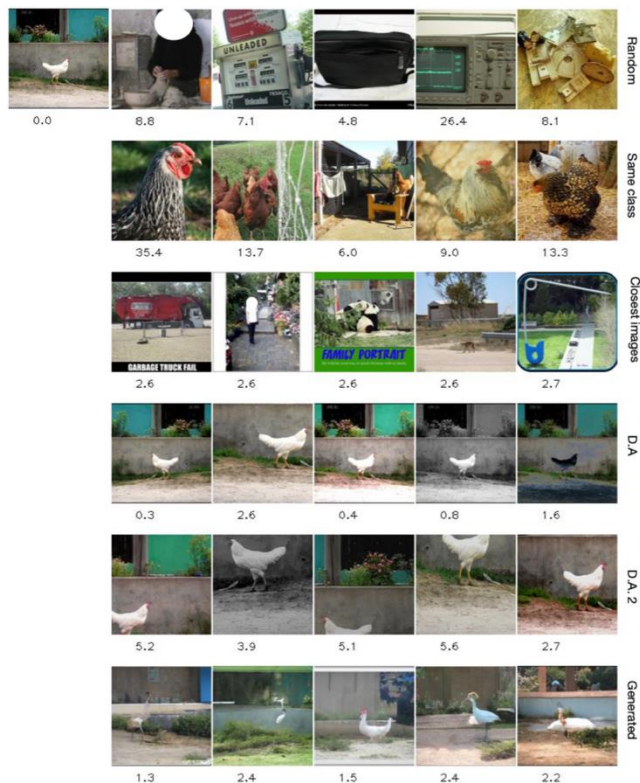


Figure 22: **Squared Euclidean distances in the SimCLR projector head representation space.** We show the squared euclidean distance between the conditioning image on the leftmost column on first row and different images to get an insight about how close the samples generated by the diffusion model stay close to the representation used as conditioning. The distances with the conditioning is printed below each images. On the first row, we show random images from the ImageNet validation data. On the second row, we take random validation examples belonging to the same class as the conditioning. On third row, we find the closest training neighbors of the conditioning in the representation space. On forth row, D.A. means Data Augmentation which consist in horizontal flip, CenterCrop, ColorJitter, GrayScale and solarization. On fifth row (D.A. 2), we use the random data Augmentation used in the paper of (Caron et al., 2020; 2021). On the last row, we show the generated samples from our conditional diffusion model that use **SimCLR projector head representation**. The samples produces by our model are much closer to the conditioning than other images.

## - Manipulation

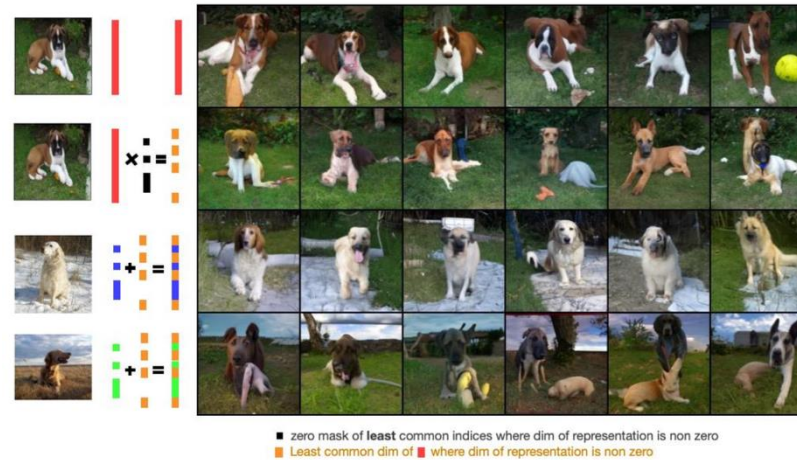


Figure 31: Same setup as Figure 30 except that instead of using the most common non zero-dimensions as mask, we used the least common non-zero dimensions as mask. On the second row, we observe that some information about the original dog is removed such that in each column, we get a slightly different breed of dog while the background stay fixed. On the third and forth row, we saw that the information about the background (grass) is propagated through the samples (which was not the case in Figure 30).

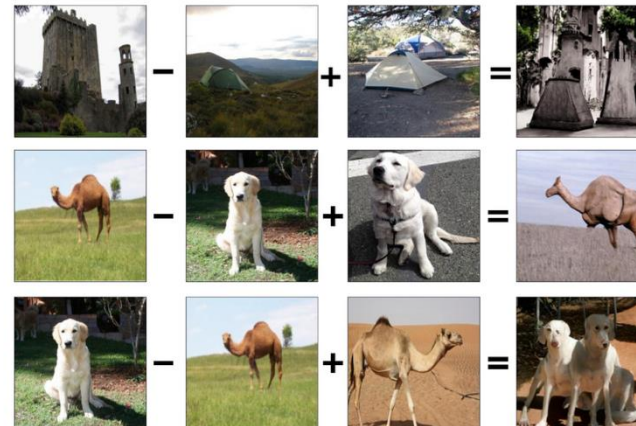


Figure 32: Algebraic manipulation of representations from real images (left-hand side of  $=$ ) allows RCDM to generate new images with novel combination of factors. Here we use this technique with ImageNet images, to attempt background substitutions.